SUNGJAE AHN

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RESEARCH INTERESTS

My conception of a fully realized AI is an agent that continually learns and adapts as it accumulates experience in a world bigger than itself. To do so, it must build on that experience to quickly identify effective actions and make informed decisions about how to modify its own behavior or structure. This goal motivates my interest in **Continual Reinforcement Learning**, **Exploration**, and **Meta-learning for RL**.

EDUCATION

Kyung Hee University (KHU) Expected Graduation: Aug. 2027
B.S. in Computer Science and Engineering
GPA: 4.31/4.5 (3.88/4.0); Upper-division: 4.48/4.5 (4.0/4.0)
Leave of absence for military service: Feb. 2022 – Nov. 2023

RESEARCH EXPERIENCE

AI & Robotics Lab, KHU Dec. 2025 – Present
Undergraduate Researcher (Advisor: Hyoseok Hwang)

- Demonstrated that temporal regularization alone can induce spatial smoothness through shared next states, allowing policies to retain performance by avoiding explicit spatial regularization.
- Demonstrated that catastrophic forgetting in the RND predictor network can be harnessed to promote re-exploration of long-unvisited states, allowing effective adaptation to changing environments.

PUBLICATIONS

Re-exploration by Leveraging Forgetting in Random Network Distillation
SungJae Ahn, Hyoseok Hwang
Submitted to Korea Software Congress 2026

Revisiting Temporal Regularization for Smooth Control in Deep Reinforcement Learning
SungJae Ahn, Jeong Woon Lee, Kyoleen Kwak, and Hyoseok Hwang
Submitted to ICRA 2027

PROJECTS

Mental Model Research for Robot Learning and Reasoning Present
Lab project with Korea Institute of Science and Technology (KIST)

- Trained and evaluated $\pi_{0.5}$ and several memory-augmented VLA models on tasks that include long-horizon manipulation and occlusion.
- Investigating the use of privileged graphs during training to ground VLA models' memory updates in changes to objects and non-object elements in video.

Analyzing Reasoning Diversity in Self-Distillation Spring 2026
Course project in Advanced Deep Learning

- Demonstrated that leveraging feedback from failed rollouts can encourage diverse reasoning strategies in self-distillation policy optimization.
- Analyzed the distinct roles of KL divergence directions in failed and successful rollouts by relating their mode-covering and mode-seeking behavior to exploration and exploitation.

Deepfake Detection Competition Fall 2025
Held by National Forensic Service

- Ranked 33 out of 266 teams utilizing Dual Data Alignment for robust cross-domain forgery detection.
- Managed experiment configurations with Hydra and used W&B to track how dataset selection and preprocessing choices led to each training run.

ACTIVITIES

SW Mentor: AI 3001 Advanced Deep Learning Present

- Designed programming assignments on core Transformer components.

SW Mentor: CSE 304 Algorithms Spring, Fall 2025

- Curated LeetCode problems for course exercises and supported students through online Q&A.

KHUDA (KHU Data analysis and AI club) 7th Jan. 2025 - Jul. 2025

- Built a legal-domain RAG chatbot to improve public access to Korean legal information, using Chroma vector search, Korean sentence embeddings, and MMR-based retrieval.

Signals Intelligence at Defense Security Agency Feb. 2022 - Nov. 2023

- Received awards for graduating first in my class (from a Major General) and identifying a previously unknown digital signal (from a Brigadier General).

SKILLS

AI/ML : PyTorch, JAX, Gymnasium, Isaac Lab, Slurm

English : TOEFL 5.5 (110)

AWARDS

Academic Excellence Scholarship, KHU Fall 2024 - Fall 2026